7(a)(i)	The half-life of a radioactive nuclide is the time taken for the activity of a sample to reduce to half its initial value.	[1] answer
7(a)(ii)	The decay constant is the fraction of the total number of nuclei in a sample that decay per unit time.	[1] answer
7(b)(i)	$E = \Delta m\, c^2 = 0.7060 \text{ MeV} = (0.7060)(1.6 \times 10^{-13})$ $= 1.1296 \times 10^{-13}$ J $\rightarrow \quad \Delta m = \dfrac{E}{c^2} = \dfrac{1.1296 \times 10^{-13}}{(3 \times 10^8)^2} = 1.2551 \times 10^{-30}$ kg $= \dfrac{1.2551 \times 10^{-30}}{1.66 \times 10^{-27}} = 7.5609 \times 10^{-4}\, u$ $M_N + M_n - (M_C + M_X) = 7.5609 \times 10^{-4}\, u$ $\rightarrow \; M_X = 1.00858\, u - 7.5609 \times 10^{-4}\, u = 1.007825\, u = 1.01u$	[1] Δm in kg [1] convert kg to u [1] answer
7(b)(ii)1.	Number produced $= \dfrac{7500}{14}(6.02 \times 10^{23}) = 3.2 \times 10^{26}$	[1] answer
7(b)(ii)2.	The probability of decay of the nucleus in a time of 1.0 year is the decay constant of the nucleus in that period of time. Decay constant, $\lambda = \dfrac{\ln 2}{t_{1/2}} = \dfrac{\ln 2}{5.7 \times 10^3} = 1.22 \times 10^{-4}$ year^{-1}	[1] answer
8(a)	Simple harmonic motion occurs when the acceleration of the	[1]